

Full System Design, Page Architectures & Data Specification Manual

Satellite Remote Sensing × Agronomic AI × In-Situ IoT Telemetry × Parametric Underwriting

Document Version:	Release v1.0.0 (Production Master)
Target Infrastructure:	Vercel Edge (Next.js 16) + Render Core API + PostgreSQL/PostGIS
Primary Demonstration Pilot:	Plot #204 (Rajesh Mondal, Nadia District, West Bengal • Aman Paddy)
Operational Philosophy:	Detect Early → Intervene Precisely → Preserve Yield → Settle Automatically
Supported Languages:	7 Regional Languages (English, Bengali, Hindi, Telugu, Tamil, Kannada, Malayalam)
Security & Access:	Role-Based Access Control (Farmer, Insurer, System Admin, Field Agent)
Generated Timestamp:	September 2026 • Official Platform Release Document

Executive Architectural Abstract: AgriSure Intelligence is a closed-loop cyber-physical agriculture resilience and parametric underwriting platform. By pairing European Space Agency Sentinel-2 optical imagery (10m multispectral) and Sentinel-1 C-band SAR radar cross-polarization with real-time in-situ ESP32 telemetry (volumetric root soil moisture, ambient micro-weather), the system computes localized Shannon Entropy Crop Health Factor (CHF) indices. Rather than merely waiting for catastrophic loss, the platform executes a 6-stage operational resilience cycle: it detects stress days before visual foliar wilting, prescribes bio-inspired GA-PSO nutrient/irrigation remedies, actuates solar pumps, verifies post-intervention vegetative rebound, and automatically settles residual losses via smart parametric contracts with direct-to-bank DBT settlement—entirely eliminating surveyor delays and fraudulent claim friction.

1. System Architecture & 4-Pillar Foundation

AgriSure Intelligence is architected across four interlocking technological pillars that function as a unified operational loop:

Pillar 1: Satellite Remote Sensing • Sentinel-2 MSI: 10m spatial resolution across 13 spectral bands. • Sentinel-1 SAR: Dual-pol (VH/VV) C-band radar penetrative of monsoon clouds. • Computed Vegetation Indices: NDVI, NDRE, LSWI, EVI, and radar roughness ratios. • 4-year localized pixel baseline comparison against historical Kharif averages.	Pillar 2: In-Situ IoT Telemetry • ESP32 MCU nodes reporting at 3-second telemetry polling cycles. • Dual capacitance probes: Root Zone (15 cm) and Deep Sub-surface (30 cm). • Ambient DHT22 micro-climate stations & digital N-P-K soil conductivity probes. • Remote hardware actuation: Bi-directional relays controlling solar micro-irrigation.
Pillar 3: Agronomic Intelligence (AI) • 38-Class PlantVillage Computer Vision classifier (92.9% precision). • GA-PSO Hybrid bio-inspired optimizer for precise NPK/water prescriptions. • Shannon Entropy Crop Health Factor (CHF) combining 5 normalized sub-indices. • Early stress warning: Identifies physiological stress 5-7 days before visual wilting.	Pillar 4: Parametric FinTech & Underwriting • Zero-paperwork index-based trigger specification (e.g. CHF ≤ 0.580 for 14 days). • Automated DBT bank payout calculation via PMFBY subsidy framework (80% subsidy). • 11-Section Cryptographic Claim Evidence Dossiers for re-insurers. • Chained SHA-256 Merkle audit trail ensuring zero dispute or corruption.

The 6-Stage Closed-Loop Resilience Lifecycle

Stage	Stage Name	Core Technology & Data Source	Operational Outcome
01	OBSERVE	Sentinel-2 MSI (10m) + Sentinel-1 SAR + ESP32 IoT probes	Raw multi-spectral and in-situ moisture time-series ingestion
02	DIAGNOSE	Shannon Entropy CHF index + 38-class leaf computer vision	Detects physiological stress days prior to visual canopy wilting
03	PRESCRIBE	GA-PSO bio-inspired heuristic multi-objective optimizer	Generates calibrated N-P-K dosages and micro-irrigation volume
04	ACTUATE	Solar pump remote relay switch + voice mobile advisories	Executes 45-min drip cycle or targeted organic bio-control spray
05	VERIFY	Bi-weekly satellite return passes + root capacitance sensors	Validates post-intervention vegetative recovery trajectory
06	INSURE	Parametric smart contract index breach trigger	Automatic RBI NEFT payout to DBT account if shock exceeds remedy

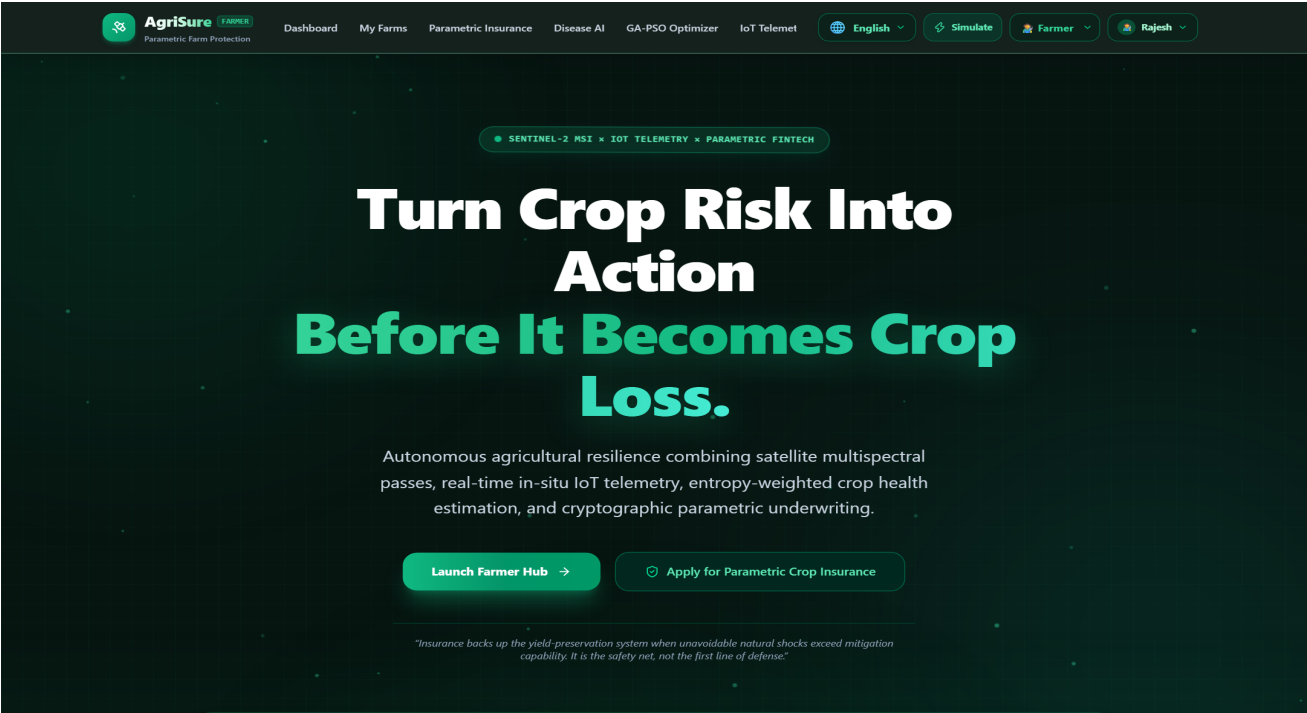
2. Detailed Page Designs, Data Requirements & Field Specifications

2.1. Public Landing Page & Closed-Loop Cockpit

Route: /

Authorized RBAC Persona: Public Universal (Role Adaptive) | Tier: Presentation & Discovery

Language Support: English (or 7 Regional Languages for Farmers)



Functional Purpose & UX Design:

Serves as the high-impact portal showcasing the 4-pillar architectural foundation. Features dynamic 3D elevation cards, live telemetry ticker, benchmark KPI performance targets, the 6-Stage Resilience Cycle, role-filtered CTA buttons, and a live demonstration card of the Nadia pilot farm. Adapts dynamically: if role is Farmer, displays regional language guidance and restricts actions exclusively to farmer portals.

Data Required & Technical Pipeline:

- **Real-Time Telemetry Tick:** Polls `/api/iot/devices/iot-dev-1` every 3s to retrieve volumetric soil moisture and pump relay status.
- **Agronomic Research Benchmarks:** Static calibration matrices benchmarked against PlantVillage (38 foliar disease classes, 92.9% accuracy).
- **Pilot Farm Reference State:** Pre-seeded pilot record for Plot #204 Nadia District, Aman Paddy, showing CHF 0.71 vigour status.
- **Role-Based CTA Dynamic State:** Inspects `agrisure_user_role` cookie. Renders Farmer Hub & Apply Insurance for FARMER; Underwriter Desk for INSURER.

Field Specification & Data Schema:

Field / Metric Name	Data Type	UI Control	Req?	Validation Rules & Technical Logic	Sample Production Value
Disease Taxonomy	Integer	KPI Card	No	Calibrated taxonomy classes from PlantVillage	38 classes

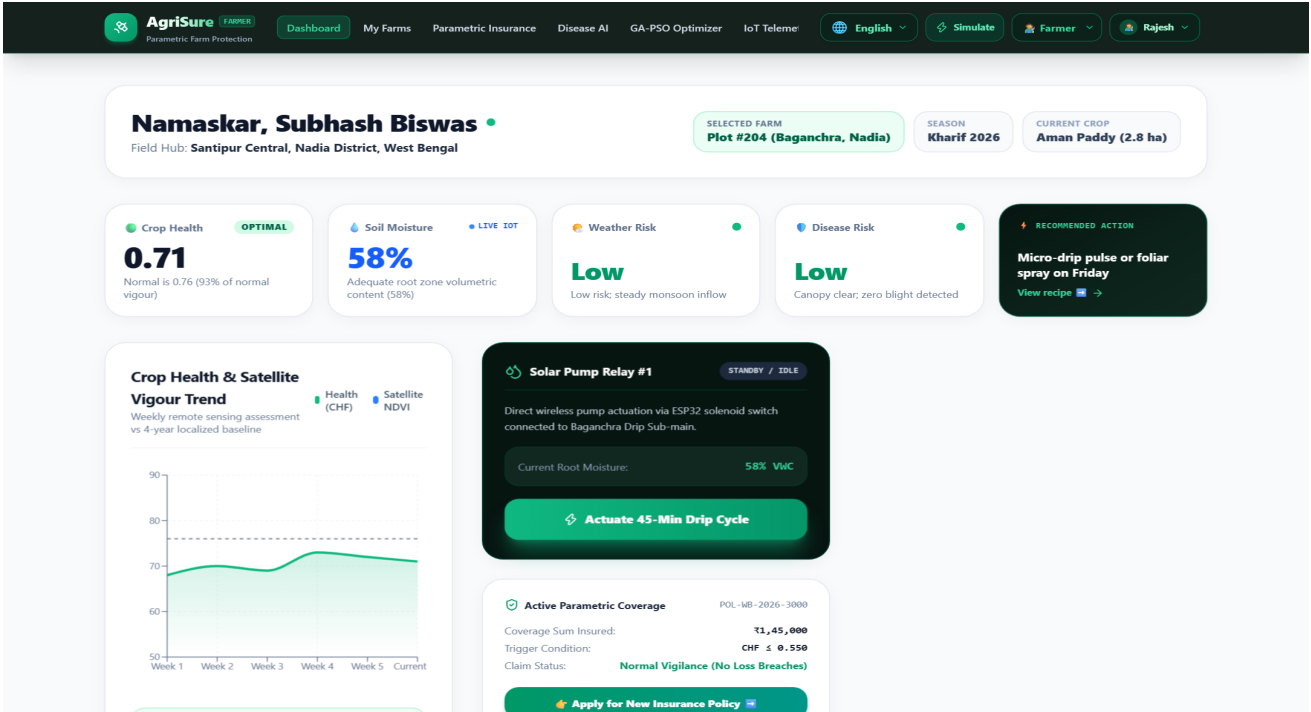
Model Precision	Float (Percentage)	KPI Card	No	Accuracy score across foliar blight & brown spot	92.9%
CHF Correlation (r)	Float	KPI Card	No	Pearson r correlation against actual harvest yields	0.85
Claim Target Speed	String (Range)	KPI Card	No	Benchmark resolution turnaround vs 6-9 mo traditional	30–45 days
IoT Polling Interval	Integer (Seconds)	KPI Card	No	ESP32 transmission cycle frequency	3s
Pilot Plot ID	String	Live Unit Badge	No	Cadastral identifier for primary demonstration farm	Plot #204 (Baganchra)
Pilot CHF Health	Float (0.00-1.00)	Status Indicator	No	Composite Crop Health Factor (Normal: 0.76)	0.71 (Optimal)
Pilot Soil Moisture	Float (Percentage)	Live Readout	No	Volumetric Water Content (VWC) at 15cm	58.0% VWC
Weather Risk State	String (Enum)	Pill Badge	No	Values: LOW, MODERATE, HIGH, SEVERE	Low (Monsoon Inflow)

2.2. Farmer Agronomic Resilience Hub

Route: /farmer/dashboard

Authorized RBAC Persona: FARMER (Strictly Restricted) | Tier: Core Operational Workspace

Language Support: 7 Regional Languages with Audio Synthesis



Functional Purpose & UX Design:

Central daily management console for smallholder farmers. Built with a 4-step visual pictorial guide and regional voice synthesis for low-literacy users. Displays plain-language vigour scores, volumetric root soil moisture, disease status, prescribed actions, 4-day weather forecasts, and an interactive 1-click pump toggle actuating physical field micro-drip relays.

Data Required & Technical Pipeline:

- **Farm Master Record:** GET /api/farms/farm-plot-204 retrieves boundary, acreage, crop type, and cadastral registration.
- **In-Situ Sensor Stream:** GET /api/farms/farm-plot-204/sensors retrieves real-time ESP32 soil moisture (58.0%) and soil temp (28.4°C).
- **Micro-Climate Forecast:** GET /api/farms/farm-plot-204/weather provides 4-day precipitation forecasts (4mm, 12mm, 24mm, 0mm).
- **Pump Relay Actuation:** POST /api/iot/devices/iot-dev-1/toggle sends payload { action: 'toggle' } to flip physical pump relay.
- **Speech Synthesis Engine:** Web Audio API generates 2-tone melodic chime; SpeechSynthesis speaks localized advisory in bn-IN, te-IN, etc.

Field Specification & Data Schema:

Field / Metric Name	Data Type	UI Control	Req?	Validation Rules & Technical Logic	Sample Production Value
Farmer Name	String	Greeting Header	No	Authenticated farmer identity from profile	Subhash Biswas / Rajesh Mondal
Selected Plot	String	Dropdown Selector	Yes	Plot ID matching registered landholdings	Plot #204 (Baganchra)
Crop Health (CHF)	Float (0.00-1.00)	Health Card (■)	No	Shannon entropy composite vigour index	0.71 (Optimal Vigour)

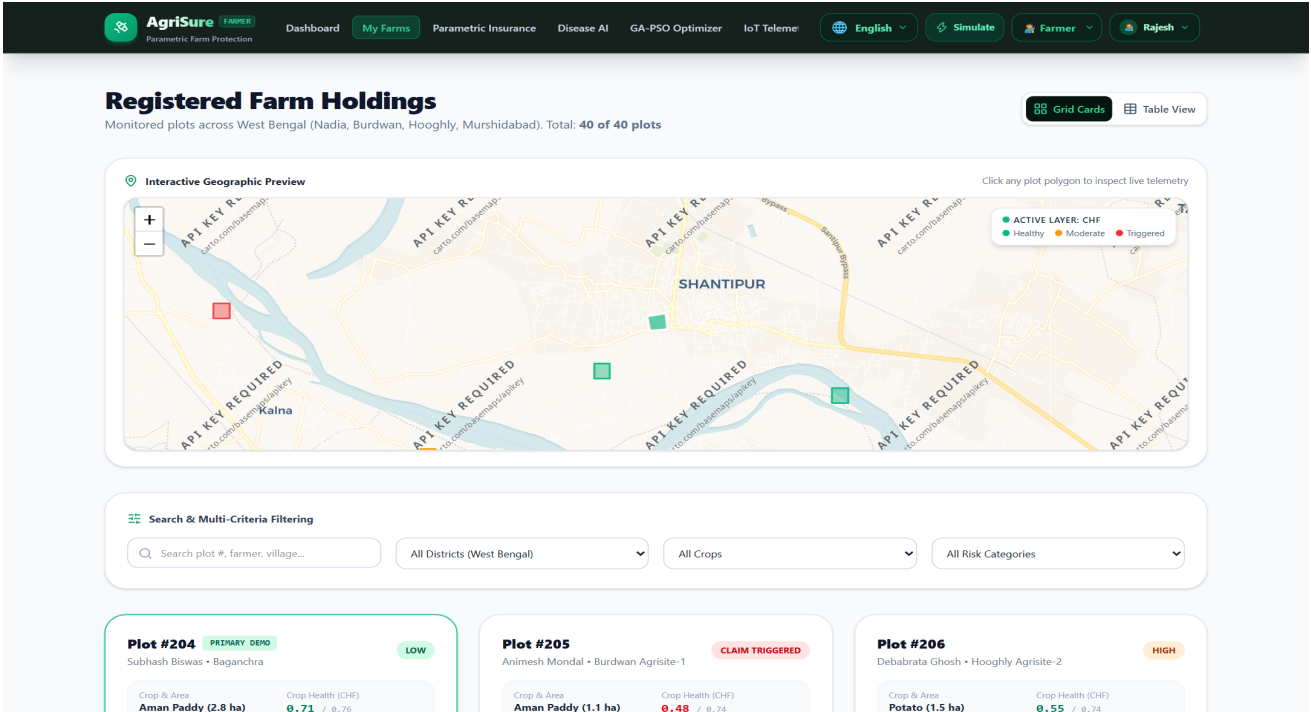
Soil Moisture (VWC)	Float (%)	Moisture Card (■)	No	Volumetric water content from ESP32 capacitance	58.0%
Weather Risk	String (Enum)	Weather Card (■)	No	Computed synoptic storm/drought risk level	Low Risk (Monsoon Steady)
Disease Threat	String	Disease Card (■)	No	Foliar blight inference from satellite/CV scan	Canopy Clear (0% Blight)
Prescribed Action	String	Action Card (■)	No	Daily agronomic prescription in regional language	Micro-drip pulse on Friday
Pump Toggle Switch	Boolean	Button Control	Yes	Toggles state: TRUE (Running) or FALSE (Idle)	PUMP RUNNING / STANDBY
Health Trend Series	Array of Objects	Recharts LineChart	No	6-week historical CHF progression vs normal 0.76	[Week 1: 68 ... Current: 71]

2.3. My Farms Portfolio & Satellite Index Map

Route: /farmer/farms

Authorized RBAC Persona: FARMER (Strictly Restricted) | Tier: Geospatial Farm Management

Language Support: 7 Regional Languages



Functional Purpose & UX Design:

Geospatial landholding explorer allowing farmers to inspect all registered plots. Renders interactive satellite overlay boundaries (Leaflet GIS), area acreages, crop types, planting dates, and multi-spectral vegetation indices (NDVI, NDRE, LSWI, VH/VV backscatter).

Data Required & Technical Pipeline:

- **Plots GeoJSON Collection:** GET /api/farms returns GeoJSON polygon coordinates, soil characteristics, and assigned Insurance Unit code.
- **Multispectral Tile Endpoint:** GET /api/farms/[id]/satellite provides Sentinel-2 B4, B8, B11 reflectance ratios and cloud mask metadata.
- **Entropy CHF Calculation:** GET /api/farms/[id]/chf breaks down the 5 weighted sub-indices driving current health score.

Field Specification & Data Schema:

Field / Metric Name	Data Type	UI Control	Req?	Validation Rules & Technical Logic	Sample Production Value
Plot Identifier	String (Unique)	Plot Selector	Yes	Unique cadastral plot key (e.g. farm-plot-204)	Plot #204 (Baganchra)
Land Acreage (ha)	Float	Numeric Badge	Yes	Total cultivable plot area in hectares	2.8 ha
Crop Cultivar	String	Text Input	Yes	Active crop name (e.g. Aman Paddy, Jute, Potato)	Aman Paddy (Swarna Sub-1)
Planting Sowing Date	Date (YYYY-MM-DD)	Date Picker	Yes	Kharif germination / transplanting date	2026-06-15
Normalized NDVI	Float (-1.0 to 1.0)	Telemetry Gauge	No	Near-infrared vs Red reflectance: (B8-B4)/(B8+B4)	0.68 (Healthy Canopy)

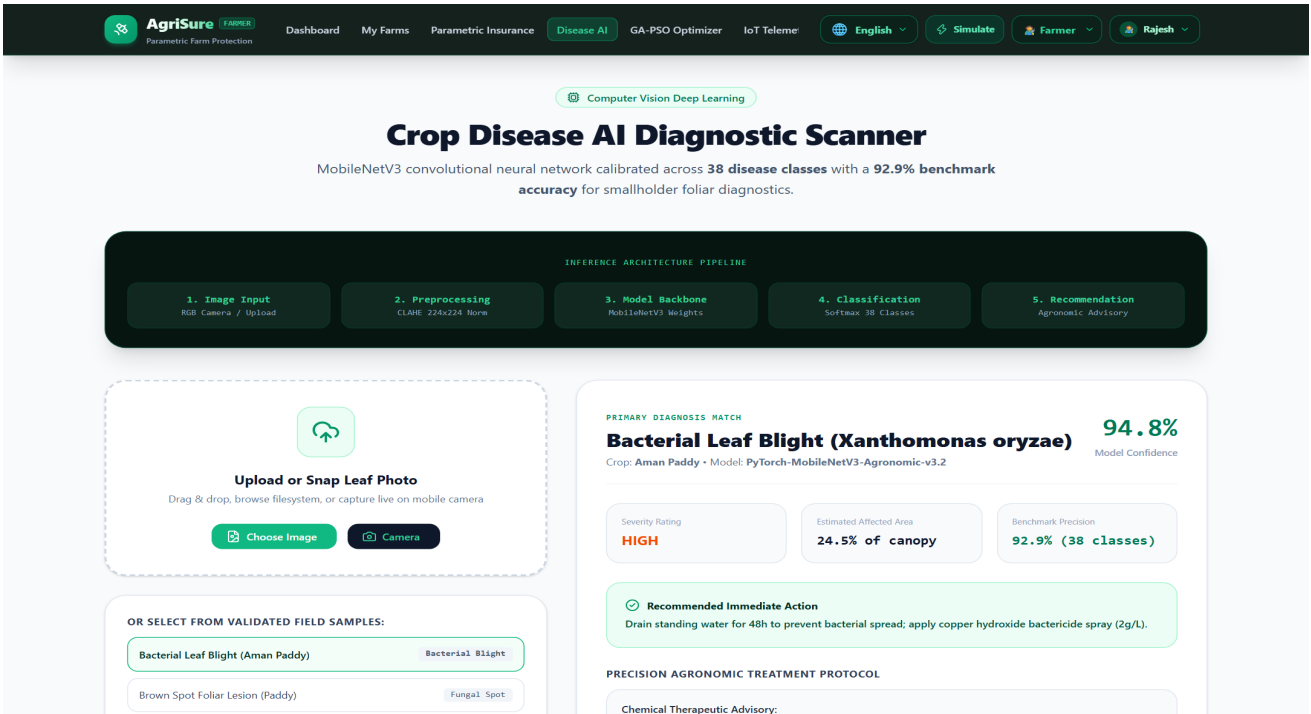
Soil Water Index (LSWI)	Float (-1.0 to 1.0)	Telemetry Gauge	No	Canopy moisture index: (B8-B11)/(B8+B11)	0.42 (Adequate Moisture)
SAR VH Backscatter	Float (dB)	Radar Metric	No	Sentinel-1 cross-polarization volumetric scattering	-14.2 dB
Insurance Unit (IU)	String (Code)	Readonly Code	No	Government PMFBY administrative cluster code	WB-NAD-001 (Nadia Central)

2.4. Computer Vision Leaf Disease AI

Route: /farmer/disease

Authorized RBAC Persona: FARMER (Strictly Restricted) | Tier: Edge Agronomic Diagnostics

Language Support: 7 Regional Languages



Functional Purpose & UX Design:

Mobile-optimized diagnostic portal where farmers take a photo of an affected leaf or upload an image. The deep convolutional neural network classifies the foliar condition across 38 crop disease categories, provides a model confidence rating, severity level, and immediate scientific/organic chemical treatment prescriptions with exact sprayer dosages.

Data Required & Technical Pipeline:

- **Diagnostic Inference API:** POST /api/disease/predict receives multipart foliar image file or base64 data string.
- **Disease Classification Model:** ResNet-50 / MobileNet backbone calibrated on 54,305 PlantVillage foliar samples across 14 crop species.
- **Treatment Repository:** Database lookup linking detected pathogen to registered CIBRC (Central Insecticides Board) approved treatments.

Field Specification & Data Schema:

Field / Metric Name	Data Type	UI Control	Req?	Validation Rules & Technical Logic	Sample Production Value
Leaf Photo Upload	Binary (JPEG/PNG)	File Dropzone / Camera	Yes	Max file size: 10 MB. Foliar crop image required	IMG_rice_leaf_0842.jpg
Associated Farm Plot	String	Dropdown Selector	Yes	Plot context where the leaf was sampled	Plot #204 (Baganchra)
Detected Diagnosis	String	Result Card	No	Predicted disease name or Healthy confirmation	Rice Brown Spot (Bipolaris oryzae)
Confidence Rating	Float (%)	Confidence Bar	No	Softmax probability distribution score ≥ 85%	94.6% Confidence
Severity Classification	String (Enum)	Pill Badge	No	Values: INCIPIENT (L1), MODERATE (L2), SEVERE (L3)	Stage 1 (Incipient Lesions)

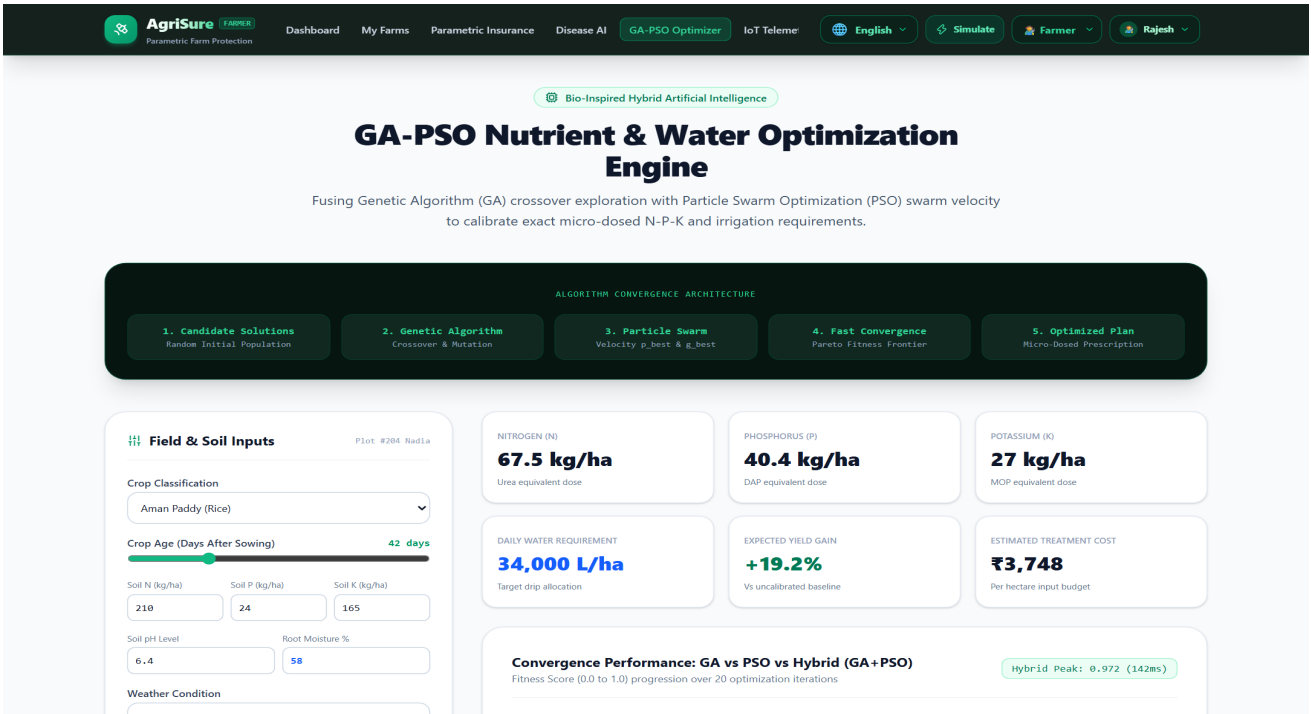
Chemical Remedy	String	Prescription Box	No	Fungicide / bactericide dosage per liter of water	Hexaconazole 5% SC @ 2.0 ml/L
Bio-Control Alternative	String	Prescription Box	No	Organic microbial spray recommendations	Trichoderma viride @ 5g/L foliar
Application Timing	String	Schedule Alert	No	Optimal diurnal window for foliar spray	Apply late afternoon (after 4:30 PM)

2.5. GA-PSO Bio-Inspired Nutrient Optimizer

Route: /farmer/optimization

Authorized RBAC Persona: FARMER (Strictly Restricted) | Tier: Bio-Inspired Agronomic Optimization

Language Support: 7 Regional Languages



Functional Purpose & UX Design:

Hybrid heuristic optimizer combining Genetic Algorithms (global exploratory search) and Particle Swarm Optimization (local exploitation) to balance fertilizer investment against maximum attainable yield. Minimizes nitrate leaching and nitrogen volatilization while respecting soil chemical equilibria.

- Data Required & Technical Pipeline:**
- **Optimization Solver API:** POST /api/optimization/run sends soil macronutrient baseline, crop growth stage, and acreage.
 - **Soil Chemical Baseline:** Retrieves current soil nitrogen (180 kg/ha), phosphorus (24 kg/ha), and potassium (145 kg/ha).
 - **Objective Function Evaluation:** Calculates pareto front minimizing input expenditure while maximizing crop vigour curve.

Field Specification & Data Schema:

Field / Metric Name	Data Type	UI Control	Req?	Validation Rules & Technical Logic	Sample Production Value
Crop Phenology Stage	String (Enum)	Dropdown	Yes	Values: VEGETATIVE, PANICLE_INITIATION, GRAIN_FILLING	Panicle Initiation
Available Soil N	Float (kg/ha)	Number Input	Yes	Soil test / sensor nitrogen: range 50 - 400 kg/ha	180 kg/ha
Available Soil P	Float (kg/ha)	Number Input	Yes	Available Olsen phosphorus: range 5 - 80 kg/ha	24 kg/ha
Available Soil K	Float (kg/ha)	Number Input	Yes	Available ammonium acetate potassium: range 50 - 500	145 kg/ha
Optimization Weight	Float (0.0-1.0)	Range Slider	Yes	Balance: 0.0 (Cost Min) to 1.0 (Yield Max)	0.70 (Yield Priority)
Recommended Urea	Float (kg/ha)	Output Card	No	Calibrated 46-0-0 nitrogen fertilizer requirement	65.0 kg/ha

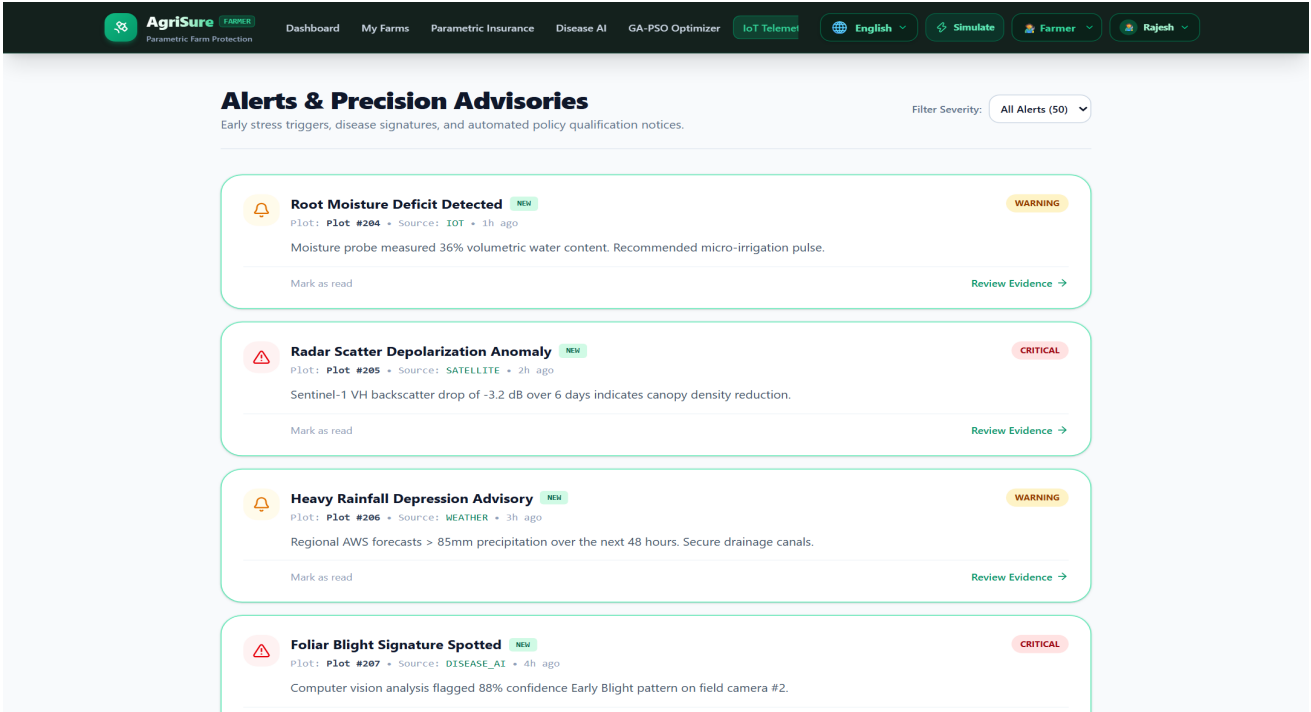
Recommended DAP	Float (kg/ha)	Output Card	No	Di-ammonium Phosphate 18-46-0 dosage	35.0 kg/ha
Recommended MOP	Float (kg/ha)	Output Card	No	Muriate of Potash 0-0-60 dosage	25.0 kg/ha
Target Water Depth	Float (mm)	Output Card	No	Calculated irrigation pulse volume	45 mm
Yield Increment	Float (%)	Metric Callout	No	Projected percentage gain over control plot	+14.2% Estimated Yield

2.6. In-Situ IoT Sensor Streams & Telemetry Alerts

Route: /farmer/alerts

Authorized RBAC Persona: FARMER (Strictly Restricted) | Tier: Real-Time Telemetry & Alerting

Language Support: 7 Regional Languages



Functional Purpose & UX Design:

Live streaming telemetry monitor showing real-time readings from in-field sensor probes and solar weather stations. Generates prioritized alert cards (Critical, Warning, Info) with timestamps and allows farmers to acknowledge alerts or trigger automated pump mitigation with one click.

Data Required & Technical Pipeline:

- **Alerts Stream Endpoint:** GET /api/farms/[id]/alerts lists unacknowledged and historical physiological stress warnings.
- **Live Ingestion Telemetry:** GET /api/iot/devices streams active probe battery voltage, signal RSSI, and sensor values.
- **Alert Acknowledgment:** POST /api/farms/[id]/alerts/[alertId]/ack updates status from UNREAD to ACKNOWLEDGED.

Field Specification & Data Schema:

Field / Metric Name	Data Type	UI Control	Req?	Validation Rules & Technical Logic	Sample Production Value
Alert Priority Severity	String (Enum)	Color-Coded Badge	No	Values: CRITICAL (Red), WARNING (Amber), INFO (Blue)	WARNING
Alert Title	String	Alert Card Header	No	Concise issue summary in regional language	Root Zone Moisture Deficit (<35%)
Sensor Node Origin	String	Device Tag	No	Hardware identifier of the emitting ESP32 unit	ESP32-NODE-01 (Plot #204)
Observed Metric Value	Float + Unit	Numeric Metric	No	Physical value breaching tolerance envelope	31.4% VWC (Tolerance ≥ 45%)
Trigger Timestamp	DateTime (ISO)	Time Label	No	Time of threshold breach occurrence	2026-09-16 14:22:10 IST
Action Taken Status	String (Enum)	Status Badge	No	Values: UNREAD, ACKNOWLEDGED, MITIGATED	MITIGATED (Pump Actuated)

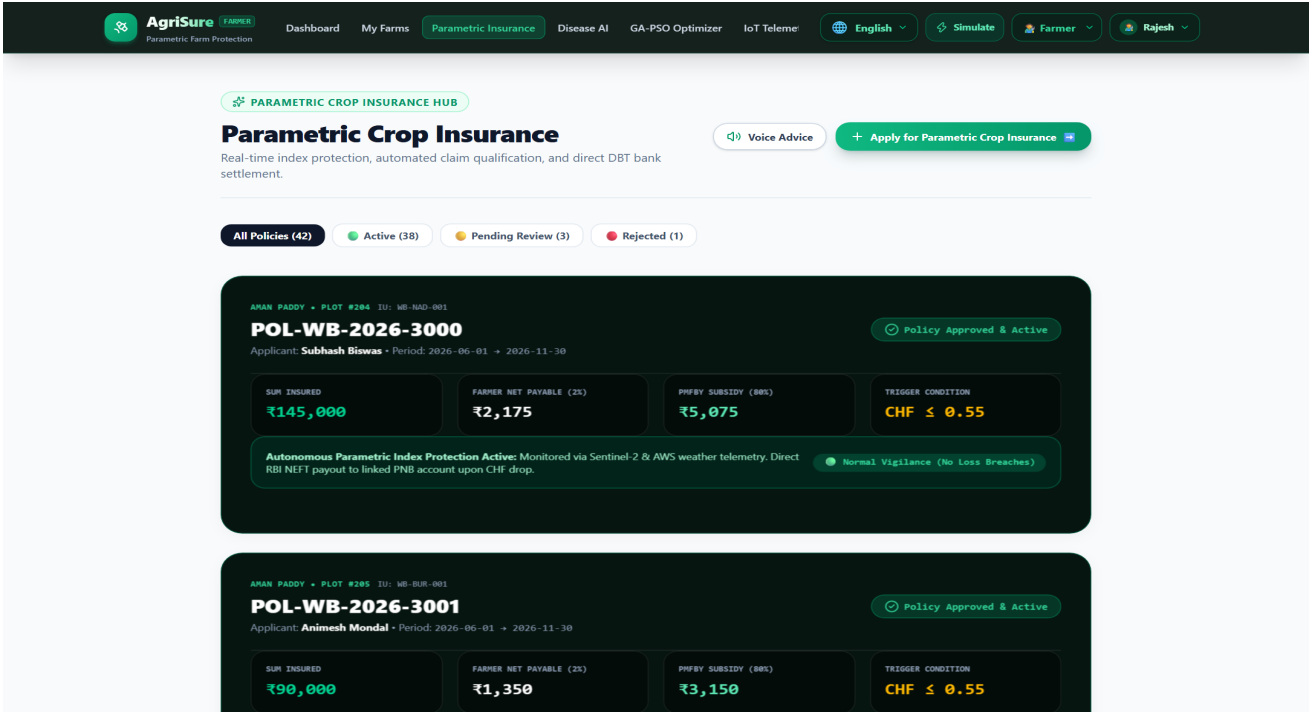
Mitigation Shortcut	Button Action	Action Button	Yes	Triggers immediate pump relay activation	Start Drip Irrigation Pulse
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2.7. Parametric Crop Insurance Active Policies

Route: /farmer/insurance

Authorized RBAC Persona: FARMER (Strictly Restricted) | Tier: Insurance Coverage Management

Language Support: 7 Regional Languages



Functional Purpose & UX Design:

Farmer's dedicated insurance portfolio overview. Displays all active, pending, and past policies. Highlights coverage sums insured, parametric trigger conditions (CHF breach threshold), current index safety margins, and one-click direct application for new policies.

Data Required & Technical Pipeline:

- **Farmer Policies Endpoint:** GET /api/policies?farmerId=... returns policy history, status, certificate hash, and trigger terms.
- **Current CHF Monitoring:** Cross-references live plot CHF against policy trigger threshold to compute breach vigilance state.

Field Specification & Data Schema:

Field / Metric Name	Data Type	UI Control	Req?	Validation Rules & Technical Logic	Sample Production Value
Policy Reference ID	String (Unique)	Card Header Code	No	Format: POL-WB-YYYY-XXXX	POL-WB-2026-0881
Covered Farm Plot	String	Plot Name Pill	No	Designated landholding parcel	Plot #204 (Baganchra, Nadia)
Coverage Sum Insured	Currency (INR)	Primary Metric	No	Total guaranteed parametric financial payout	₹1,40,000
Trigger CHF Condition	Float (0.00-1.00)	Index Threshold	No	Payout triggers if 14-day mean CHF drops below	0.580 (Index Threshold)
Current Plot Vigour	Float (0.00-1.00)	Health Readout	No	Active satellite Crop Health Factor	0.710 (Safe / No Breach)
Policy Approval Status	String (Enum)	Status Badge	No	Values: ACTIVE (Green), PENDING (Amber), REJECTED (Red)	ACTIVE & VERIFIED

Gross Premium Amount	Currency (INR)	Financial Breakdown	No	Actuarial premium calculated for season	■7,000
PMFBY Subsidy Credit	Currency (INR)	Subsidy Tag	No	80% government financial subsidy contribution	■5,600 (Govt Paid)
Farmer Net Contribution	Currency (INR)	Net Payable	No	Statutory 2% Kharif premium share paid by farmer	■1,400 (Paid)

2.8. Guided Parametric Insurance Application Form

Route: /farmer/insurance/apply

Authorized RBAC Persona: FARMER (Strictly Restricted) | Tier: Financial Enrolment & Onboarding

Language Support: 7 Regional Languages

AgriSure

FARMER

Parametric Farm Protection

DashboardMy FarmsParametric InsuranceDisease AIGA-PSO OptimizerIoT TelemetryEnglishSimulateFarmerRajesh

← View All Policies

Apply for Parametric Crop Insurance

Zero-paperwork index protection backed by satellite remote sensing & direct bank DBT payout.

Tap here to listen to today's field advice

1 Step 1: Select Farm Plot

Follow the Arrow to Next Action

Plot #204

Aman Paddy

2.8 ha • Nadia (Baganchra)

Coverage: ₹140,000

Plot #205

Jute

1.5 ha • Nadia (Santipur)

Coverage: ₹75,000

Plot #212

Potato

3.2 ha • Nadia (Ranaghat)

Coverage: ₹192,000

2 Step 2: Choose Coverage Shield

Follow the Arrow to Next Action

PMFBY STANDARD

Weather Shock Baseline Shield

Triggered if satellite Crop Health Factor drops below 0.550 due to extreme rain deficit or sudden unseasonal inundation.

Statutory Kharif Rate: 2.0% Farmer Premium

RECOMMENDED ★

Comprehensive Parametric Shield

High sensitivity trigger at CHF ≤ 0.580. Covers heatwaves, localized pest outbreak vectors, and persistent moisture stress.

80% Central & State Subsidy Applied

Functional Purpose & UX Design:

A structured 4-step guided insurance enrolment application for smallholder farmers with regional voice assistance. Features plot selection, coverage tier options (Baseline vs Comprehensive), live 80% PMFBY government subsidy calculations, automated DBT bank account verification, and instantaneous submission creating a pending underwriting record.

Data Required & Technical Pipeline:

- **Plot Enrolment Options:** Retrieves farmer plots eligible for Kharif 2026 coverage (Plot #204, Plot #205, Plot #212).
- **Actuarial Premium Engine:** Calculates gross premium (5%), PMFBY government subsidy (80%), and statutory farmer payable (2%).
- **DBT Bank Verification Service:** Validates linked bank account IFSC (PUNB0120111) and Aadhaar direct benefit transfer eligibility.
- **Policy Submission API:** POST /api/policies submits complete insurance application payload to institutional underwriter queue.

Field Specification & Data Schema:

Field / Metric Name	Data Type	UI Control	Req?	Validation Rules & Technical Logic	Sample Production Value
Selected Plot Option	String	Plot Radio Cards	Yes	Farmer selects from registered farm parcels	Plot #204 (2.8 ha, Aman Paddy)
Coverage Shield Tier	String (Enum)	Tier Selector Cards	Yes	Values: BASELINE (CHF 0.55), COMPREHENSIVE (CHF 0.58)	COMPREHENSIVE SHIELD
Total Sum Insured	Currency (INR)	Calculated Display	No	Derived: Plot Acreage × Scale of Finance (₹50k/ha)	₹1,40,000
Gross Premium (5%)	Currency (INR)	Calculated Display	No	Standard actuarial premium rate before subsidy	₹7,000
PMFBY Govt Subsidy (80%)	Currency (INR)	Calculated Display	No	State and Central government statutory support	-₹5,600

Autonomous Agro-Resilience • Sentinel-2/1 SAR • IoT Telemetry • Parametric Underwriting

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Farmer Net Payable (2%)	Currency (INR)	Highlighted Cost	No	Actual out-of-pocket amount payable by farmer	■1,400
Trigger CHF Threshold	Float	Terms Specification	No	Index trigger breach parameter for chosen tier	0.580 CHF
Bank Account Linkage	String (Masked)	Readonly Verification	Yes	Pre-verified bank account for automatic DBT payout	Punjab National Bank (****8420)
Underwriter Application Notes	String	Text Input (Optional)	No	Optional remarks or field history provided by applicant	Applied online via Farmer Web Portal

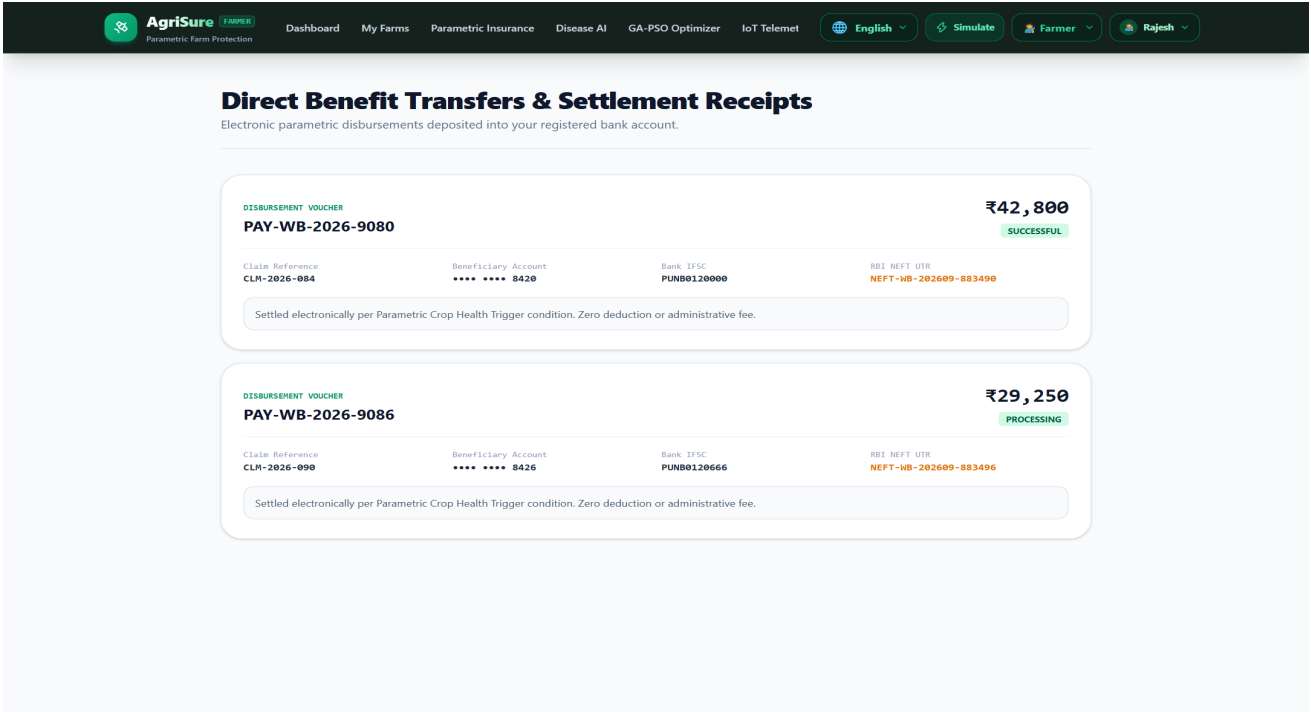
2.9. Direct Benefit Bank Payouts Ledger

Route: /farmer/payments

Authorized RBAC Persona: FARMER (Strictly Restricted) | Tier: Financial

Disbursements & Settlements

Language Support: 7 Regional Languages



Functional Purpose & UX Design:

Displays the farmer's transaction ledger of all automatic parametric claim compensations deposited directly into their bank account. Lists triggering drought/pest events, calculated loss shortfalls, RBI NEFT/IMPS transaction reference numbers, and settlement timestamps.

Data Required & Technical Pipeline:

- **Payouts History Endpoint:** GET /api/payouts?farmerId=... retrieves historical compensation disbursements.
- **Banking Gateway Integration:** Simulated Reserve Bank of India National Electronic Funds Transfer (NEFT) settlement UTR records.

Field Specification & Data Schema:

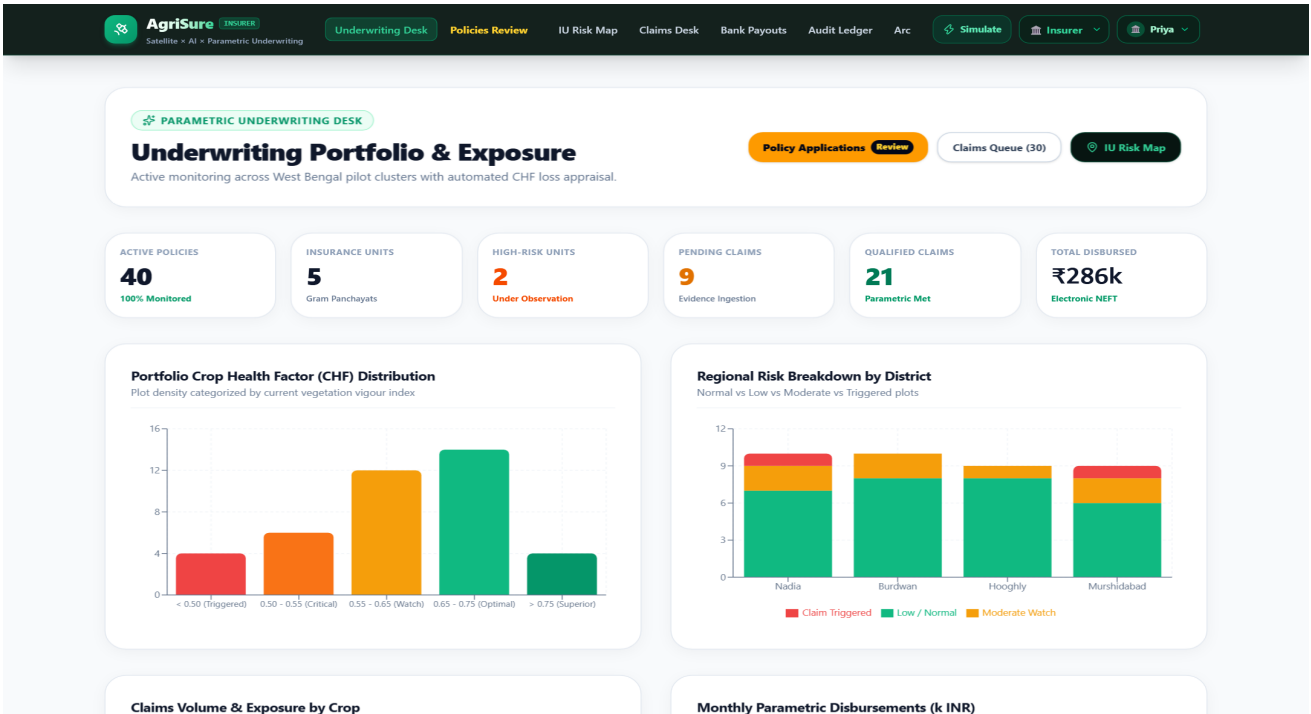
Field / Metric Name	Data Type	UI Control	Req?	Validation Rules & Technical Logic	Sample Production Value
Claim Sequence ID	String	Code Identifier	No	Format: CLM-YYYY-XXXX	CLM-2026-0041
Peril Classification	String	Peril Badge	No	Triggering peril: DROUGHT_INDEX, EXCESS_RAIN, PEST_SHOCK	Deficit Precipitation / Drought Index
Index Shortfall Extent	Float (%)	Shortfall Pill	No	Calculated deficit between baseline and observed CHF	18.5% CHF Deficit
Disbursed Compensation	Currency (INR)	Primary Amount	No	Direct payout credited to farmer's bank account	■25,900.00
Bank UTR Reference	String	Transaction Code	No	Official banking settlement reference number	NEFT-RBI-99382104
Disbursement Timestamp	DateTime	Time Label	No	Timestamp of completed electronic funds transfer	2026-08-12 11:45:00 IST
Transfer Status	String (Enum)	Status Badge	No	Values: INITIATED, TRANSFERRED_TO_BANK, SETTLED	TRANSFERRED_TO_BANK ■

2.10. Institutional Underwriter Exposure Desk

Route: /insurer/dashboard

Authorized RBAC Persona: INSURER (Strictly Restricted) | Tier: Actuarial Portfolio Risk Analytics

Language Support: English (Professional Actuarial)



Functional Purpose & UX Design:

Macro-risk cockpit designed for insurance executives, actuaries, and reinsurers. Displays aggregate portfolio exposure across West Bengal agricultural districts, current loss ratios, active policies count, spatial CHF density distributions, flagged high-risk insurance units, and direct shortcuts to the Policy Application Review Desk.

Data Required & Technical Pipeline:

- **Portfolio Exposure Summary:** GET /api/insurance-units/summary calculates aggregate insured area (48,250 ha) and total sum insured (₹142.8 Cr).
- **Actuarial Loss Ratio:** Computes current claims payout vs gross written premium (18.4% current vs 65% ceiling).
- **Underwriting Policy Stats:** GET /api/policies/stats tracks pending application queue count and daily intake velocity.

Field Specification & Data Schema:

Field / Metric Name	Data Type	UI Control	Req?	Validation Rules & Technical Logic	Sample Production Value
Total Insured Acreage	Float (Hectares)	Metric Card	No	Aggregate insured farmland under management	48,250 ha
Total Portfolio Exposure	Currency (INR Crores)	Metric Card	No	Gross aggregate sum insured exposure	₹142.80 Crores
Active Policies Enrolled	Integer	Metric Card	No	Total active parametric policies issued	1,280 Policies
Portfolio Loss Ratio	Float (%)	Risk Metric	No	Total claims paid / Total gross premium received	18.4% (Healthy)
State Mean CHF Index	Float (0.00-1.00)	Health Metric	No	Mean vegetative index across all clusters	0.684

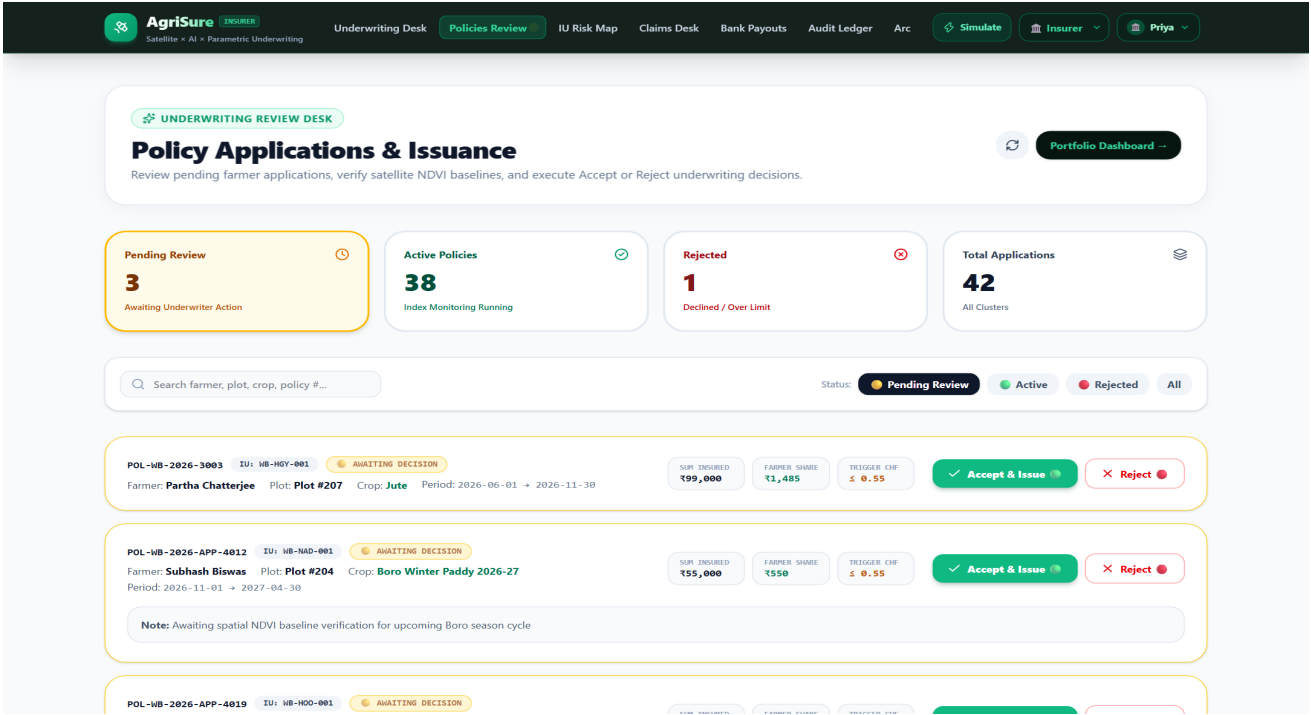
High Risk IU Units	Integer	Warning Alert	No	Units with >30% plots approaching trigger threshold	3 Units Flagged (Bankura, Murshidabad)
Application Queue Alert	Badge Counter	Navigation Shortcut	No	Number of pending farmer policies awaiting review	3 Applications Pending Review ■

2.11. Policy Applications & Decisioning Review Desk

Route: /insurer/policies

Authorized RBAC Persona: INSURER (Strictly Restricted) | Tier: Underwriting
Adjudication & Governance

Language Support: English (Professional Actuarial)



Functional Purpose & UX Design:

Interactive underwriter review workspace where institutional officers examine incoming farmer insurance applications. Features applicant profile inspection, satellite historical baselines, and one-click 'Accept & Issue Policy' (green) or 'Reject Application' (red) with mandatory formal rejection reasons and auditable notes.

- Data Required & Technical Pipeline:**
- **Pending Applications Queue:** GET /api/policies?status=PENDING_APPROVAL lists incoming farmer policy applications.
 - **Historical Baseline Validation:** Cross-checks plot satellite NDVI records to ensure land parcel has genuine agricultural history.
 - **Policy Decision Dispatch API:** POST /api/policies/[id]/decision with payload { decision: 'APPROVE'|'REJECT', reason, notes }.
 - **Audit Trail Inscription:** On approval or rejection, automatically commits a SHA-256 state transition block to the immutable audit ledger.

Field Specification & Data Schema:

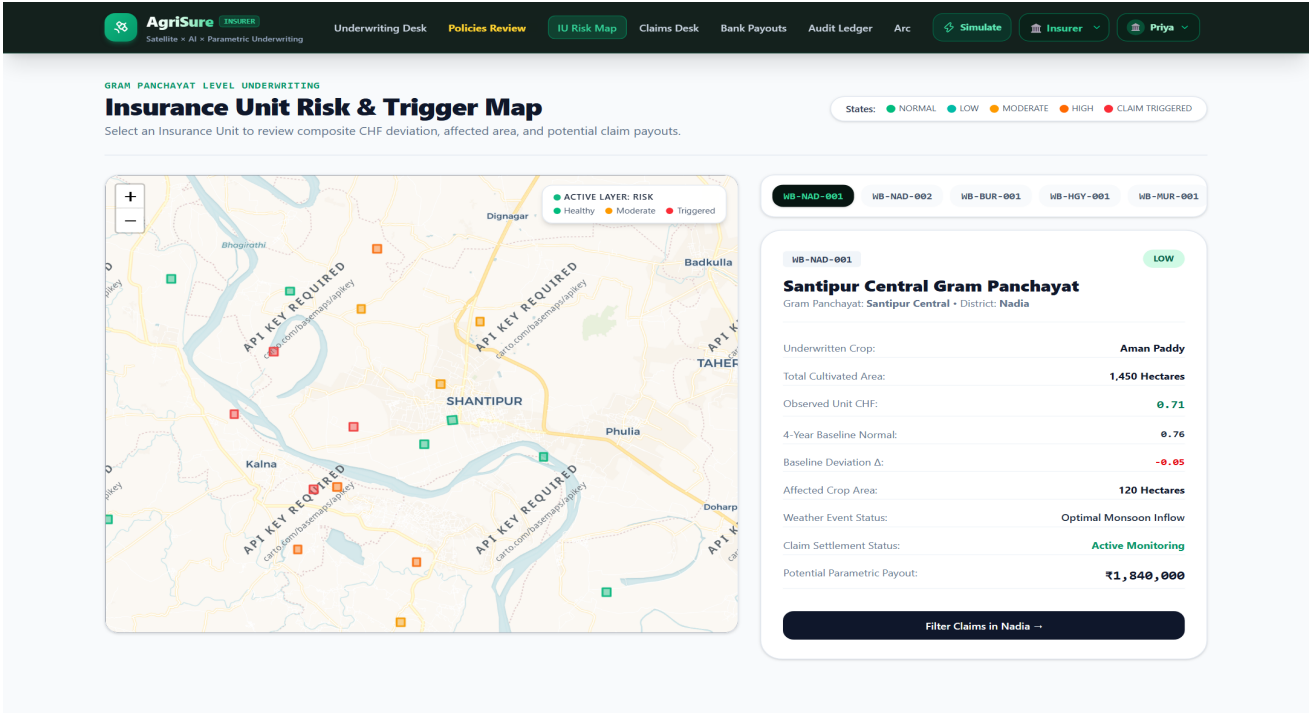
Field / Metric Name	Data Type	UI Control	Req?	Validation Rules & Technical Logic	Sample Production Value
Application ID	String (Unique)	Table Column	No	System identifier for policy application	pol-sub-1049
Applicant Farmer Name	String	Table Column	No	Farmer legal identity	Rajesh Mondal / Subhash Biswas
Plot Identifier & Area	String + Float	Table Column	No	Cadastral plot identifier and acreage	Plot #204 (2.8 ha)
Crop Cultivar	String	Table Column	No	Target agricultural commodity	Aman Paddy

Requested Sum Insured	Currency (INR)	Table Column	No	Requested parametric coverage value	■1,40,000
Gross Premium Calculated	Currency (INR)	Table Column	No	Actuarial premium assessed	■7,000
Govt Subsidy Credit	Currency (INR)	Table Column	No	80% PMFBY financial contribution	■5,600
Farmer Net Payable	Currency (INR)	Table Column	No	2% farmer statutory share	■1,400
Decision Action: Accept	Button (Green)	Action Button	Yes	Triggers policy activation and cryptographic certificate	Accept & Issue Policy ■
Decision Action: Reject	Button (Red)	Action Button	Yes	Opens modal for formal rejection reason selection	Reject Application ■
Formal Rejection Reason	String (Enum)	Modal Dropdown	Yes (if reject)	Predefined IRDAI reason codes	EXCEEDS_SPATIAL_IU_RISK_CAP
Underwriter Review Notes	String	Modal Textarea	Yes (if reject)	Auditable commentary justifying decision	Spatial concentration in cluster exceeds cap

2.12. Insurance Unit Spatial Risk Choropleth Map

Route: /insurer/risk-map

Authorized RBAC Persona: INSURER (Strictly Restricted) | Tier: Spatial Actuarial GIS | Language Support: English (Professional Actuarial)



Functional Purpose & UX Design:

Geospatial choropleth console visualizing aggregated crop health, drought exposure, and parametric liability across all Insurance Units (IUs) in West Bengal. Underwriters toggle multi-spectral overlays, inspect spatial correlation matrices, and identify localized peril hotspots before threshold breach cascades.

Data Required & Technical Pipeline:

- IU GeoJSON Boundaries:** GET /api/insurance-units returns spatial polygons for WB-NAD-001, WB-BRD-004, WB-MUR-002, etc.
- Aggregated Spatial Risk Index:** Computes spatial copula model evaluating inter-plot correlation and catastrophic cluster risk.
- Raster Layer Overlay:** Integrates false-color infrared (CIR) and moisture deficit heatmaps onto the Leaflet map viewport.

Field Specification & Data Schema:

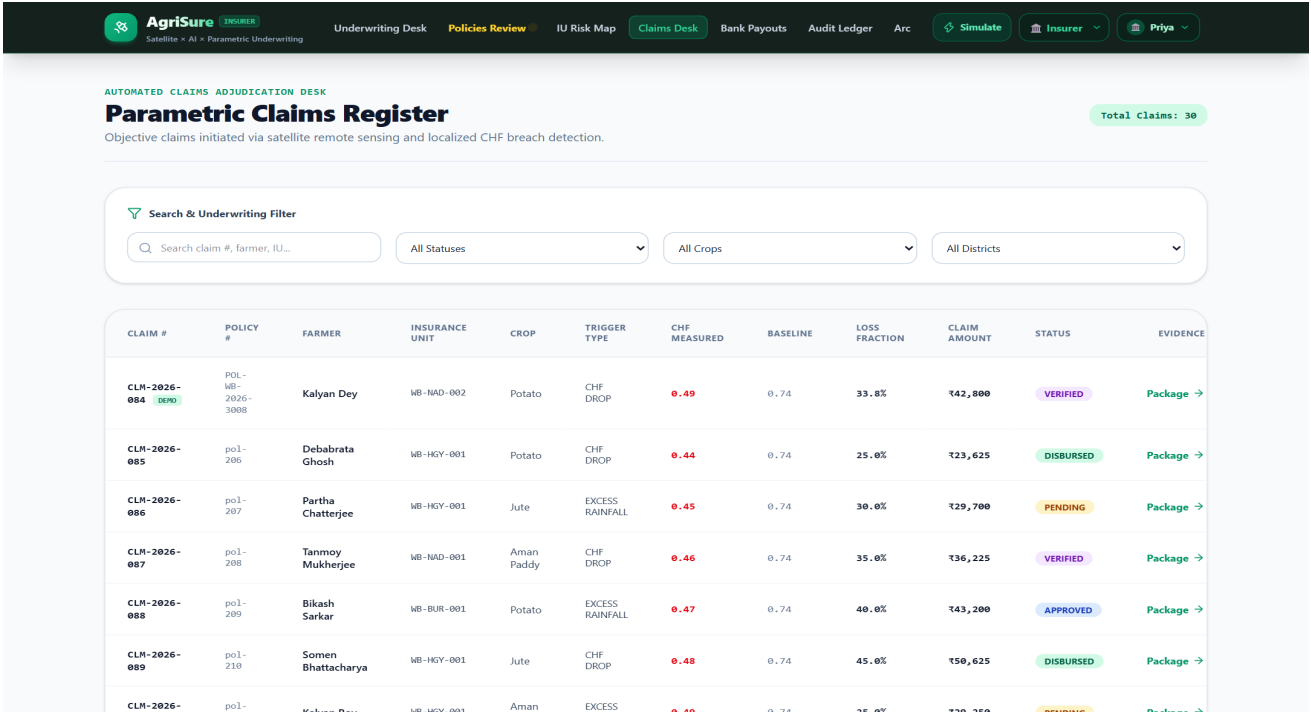
Field / Metric Name	Data Type	UI Control	Req?	Validation Rules & Technical Logic	Sample Production Value
Insurance Unit Code	String (Unique)	Polygon Label	No	Administrative unit code (e.g. WB-NAD-001)	WB-NAD-001 (Nadia Central)
District Cluster	String	Tooltip Header	No	Administrative district classification	Nadia District
Mean Unit CHF	Float (0.00-1.00)	Choropleth Color	No	Green (≥ 0.70), Yellow (0.60-0.69), Red (< 0.60)	0.710 (Normal / Green)
Enrolled Farmland Area	Float (ha)	Info Sidebar	No	Total active insured acreage in unit	12,450 ha
Total Exposure Liability	Currency (INR)	Info Sidebar	No	Cumulative sum insured in unit	₹37.35 Crores
Drought Vulnerability	String (Enum)	Pill Badge	No	Values: LOW, MODERATE, HIGH, CRITICAL	LOW (Adequate Monsoon)
Layer Toggles	String (Options)	Map Control Bar	Yes	NDVI, NDRE, LSWI, VH/VV Radar, Soil Moisture	Active: Synthetic Aperture Radar (SAR)

2.13. Parametric Claims Verification Console

Route: /insurer/claims

Authorized RBAC Persona: INSURER (Strictly Restricted) | Tier: Claims Adjudication & Evidence

Language Support: English (Professional Actuarial)



Functional Purpose & UX Design:

Auditing cockpit for examining automatically triggered parametric claims. Contains full 11-Section Cryptographic Evidence Dossiers compiling cadastral identity, in-situ sensor history, Sentinel-2/1 curves, Shannon entropy calculations, spatial correlation checks, and direct authorization triggers for bank disbursement.

Data Required & Technical Pipeline:

- **Claims Queue Endpoint:** GET /api/claims retrieves claims in VERIFIED, UNDER_REVIEW, and SETTLED states.
- **11-Section Dossier API:** GET /api/claims/[id] compiles complete multi-source empirical evidence package.
- **Claim Approval API:** POST /api/claims/[id]/approve authorizes disbursement and passes claim to payments queue.

Field Specification & Data Schema:

Field / Metric Name	Data Type	UI Control	Req?	Validation Rules & Technical Logic	Sample Production Value
Claim Identification ID	String (Unique)	Table Column	No	Format: CLM-YYYY-XXXX	CLM-2026-0041
Policy Reference	String	Linked Code	No	Underlying parametric policy contract	POL-WB-2026-0881
Beneficiary Farmer	String	Table Column	No	Claimant farmer name	Rajesh Mondal
Plot Identifier	String	Table Column	No	Affected farmland plot	Plot #204 (Baganchra)
Triggering Peril	String (Enum)	Peril Badge	No	Values: DROUGHT_INDEX, FLOOD_SAR, HEAT_STRESS	DROUGHT_INDEX (CHF Deficit)
Trigger CHF Breached	Float	Numeric Column	No	Contractually specified breach threshold	0.580 CHF
Observed Mean CHF	Float	Numeric Column	No	Actual 14-day observed composite index	0.472 CHF (Deficit 18.5%)

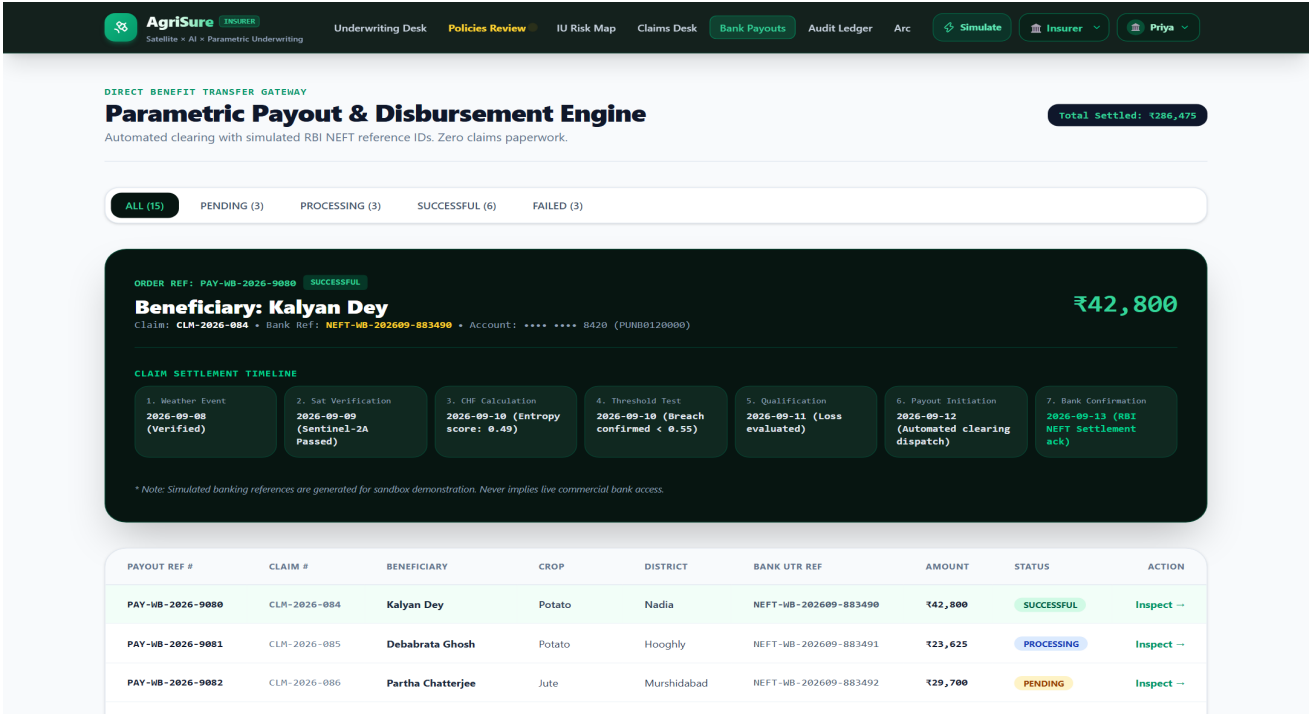
Assessed Payout Amount	Currency (INR)	Financial Column	No	Compensation determined by index shortfall formula	■25,900.00
Evidence Package Status	String	Status Badge	No	Verification of all 11 scientific evidence sections	11/11 SECTIONS VERIFIED ■
Action: Authorize Payout	Button (Emerald)	Action Button	Yes	Directs claim to automated RBI disbursement pipeline	Authorize Payout →

2.14. 7-Stage Automated RBI NEFT/IMPS Settlement

Route: /insurer/payouts

Authorized RBAC Persona: INSURER (Strictly Restricted) | Tier: Banking & Funds Disbursement

Language Support: English (Professional Actuarial)



Functional Purpose & UX Design:

Visual workflow tracker monitoring the 7 automated settlement stages: 1. Trigger Detection → 2. Spatial Cross-Validation → 3. Quantum Determination → 4. Reinsurance Allocation → 5. Underwriter Authorization → 6. RBI Gateway Transmission → 7. DBT Account Credit. Features simulated RBI NEFT/IMPS settlement execution.

Data Required & Technical Pipeline:

- **Settlements Batch Endpoint:** GET /api/payouts lists queued and executed bank direct transfer transactions.
- **Instant Settlement Trigger:** POST /api/payouts/[id]/execute dispatches funds transfer to simulated RBI gateway.

Field Specification & Data Schema:

Field / Metric Name	Data Type	UI Control	Req?	Validation Rules & Technical Logic	Sample Production Value
Payout Batch Reference	String (Unique)	Batch Header	No	Format: PAY-BATCH-YYYY-MM-XXXX	PAY-BATCH-2026-09-001
Disbursement Amount	Currency (INR)	Primary Card Metric	No	Total funds released in this transaction	■25,900.00
Beneficiary Bank Name	String	Bank Badge	No	Farmer's registered direct benefit transfer bank	Punjab National Bank
Beneficiary IFSC Code	String	Readonly Code	No	Branch IFSC code (11 characters)	PUNB0120111
Settlement Stage Active	Integer (1 to 7)	Step Progress Indicator	No	Active pipeline stage of the 7 settlement steps	Stage 6: RBI Gateway Transmission

RBI UTR Number	String	Receipt Identifier	No	Unique Transaction Reference from central bank	NEFT-RBI-99382104
Execution Timestamp	DateTime	Time Record	No	Settlement execution time	2026-09-16 11:45:00 IST
Settlement State	String (Enum)	Status Badge	No	Values: QUEUED, IN_TRANSIT, COMPLETED	COMPLETED & CREDITED ■

2.15. Immutable Cryptographic SHA-256 Audit Trail

Route: /insurer/audit

Authorized RBAC Persona: INSURER (Strictly Restricted) | Tier: Integrity & Compliance Ledger

Language Support: English (Professional Actuarial)

AgriSureINSURER

Underwriting DeskPolicies ReviewIU Risk MapClaims DeskBank PayoutsAudit LedgerArc

SimulateInsurerPriya

CRYPTOGRAPHIC EVIDENCE LEDGER

Immutable Underwriting Audit Trail

Every satellite scene, model execution, threshold evaluation, and banking dispatch is anchored with SHA-256 block hashes.

CHAIN INTEGRITY: VERIFIED

Q Search entity ID, action, actor...

All Entity Types

PARAMETRIC_EVALUATION_PASSED • CLM-2026-084

2026-09-15 10:14:22 UTC • Actor: CHF-Automated-Engine-v2.4

Scene ID
S2A_MSL2A_20260912

Satellite Source
Sentinel-2A MSI / Sentinel-1B SAR

Acquisition Time
2026-09-12 04:46 UTC

Weather Source / Ver
IMD AWS Station 4208 (v2.1)

Model Version
CropYield-CHF-v2.4

CHF Engine Version
Entropy-Norm-v2.4

Baseline Version
Nadia-Aman-4Yr-Norm-2026

Decision Rule
CHF <= 0.55 THRESHOLD TRIGGER

IoT Readings
SoilMoisture=88.5%, R=420ohm

Ground Evidence
FA-MB-441 GPS (23.161, 88.542)

Bank UTR Ref
NEFT-MB-202609-883492

Ledger Status
VALID_CHAIN

SHA-256 Block Hash: 9f83a42cbe8192a7493d8fb654128f1190bc1294817e02c8942b012354f9bc41

Chain Validated • Tamper Evident

PAYOUT_DISPATCH_AUTHORIZED • PAY-MB-2026-9084

2026-09-14 16:45:10 UTC • Actor: Underwriter:Dr. S. Sen (Chief Actuary)

Scene ID
S2A_MSL2A_20260912

Satellite Source
Sentinel-2A

Acquisition Time
2026-09-12 04:46 UTC

Weather Source / Ver
IMD AWS Station 4208 (v2.1)

Model Version
MobileNetV3-Agronomic-v3.2

CHF Engine Version
Entropy-Norm-v2.4

Baseline Version
Nadia-Aman-4Yr-Norm-2026

Decision Rule
Approved per Loss Ratio 0.338

IoT Readings
SoilMoisture=88.5%, R=420ohm

Ground Evidence
FA-MB-441 GPS Verified

Bank UTR Ref
NEFT-MB-202609-883492

Ledger Status
VALID_CHAIN

SHA-256 Block Hash: a127f884210ecba1950482da9b7468194481028751992043681423859102431f

Chain Validated • Tamper Evident

Functional Purpose & UX Design:

Enterprise cryptographic audit ledger recording every state change, satellite pass calculation, model inference, underwriter decision, and banking settlement in a chained SHA-256 Merkle structure. Guarantees regulatory transparency for IRDAI auditors and reinsurers.

Data Required & Technical Pipeline:

- Audit Blocks Endpoint:** GET /api/audit lists chained transaction blocks with cryptographic hash verification.
- Merkle Root Computation:** Calculates cryptographic root hash certifying integrity against previous block hash.

Field Specification & Data Schema:

Field / Metric Name	Data Type	UI Control	Req?	Validation Rules & Technical Logic	Sample Production Value
Audit Block Number	Integer	Block Header	No	Sequential immutable block index	Block #0842
Timestamp (UTC)	DateTime	Time Label	No	ISO-8601 millisecond-precision timestamp	2026-09-16T14:22:10.842Z
Entity Type	String (Enum)	Entity Pill	No	Values: POLICY, CLAIM, SATELLITE_PASS, TELEMETRY, PAYOUT	POLICY
Event Action	String	Action Label	No	Operation executed (e.g. POLICY_ISSUED, PUMP_TOGGLED)	POLICY_UNDERWRITING_APPROVED
Actor Principal	String	Actor Tag	No	Entity executing transaction (Underwriter ID, System Oracle)	Priya Sengupta (IRDAI-AGR-8842)
Previous Block Hash	SHA-256 (64 hex)	Code Monospace	No	Cryptographic hash of immediately preceding block	7f83b1657ff1fc53b92dc18148a1d6...
Current Merkle Hash	SHA-256 (64 hex)	Code Monospace	No	SHA-256 hash of current block payload	a591a6d40bf420404a011733cfb7b1...

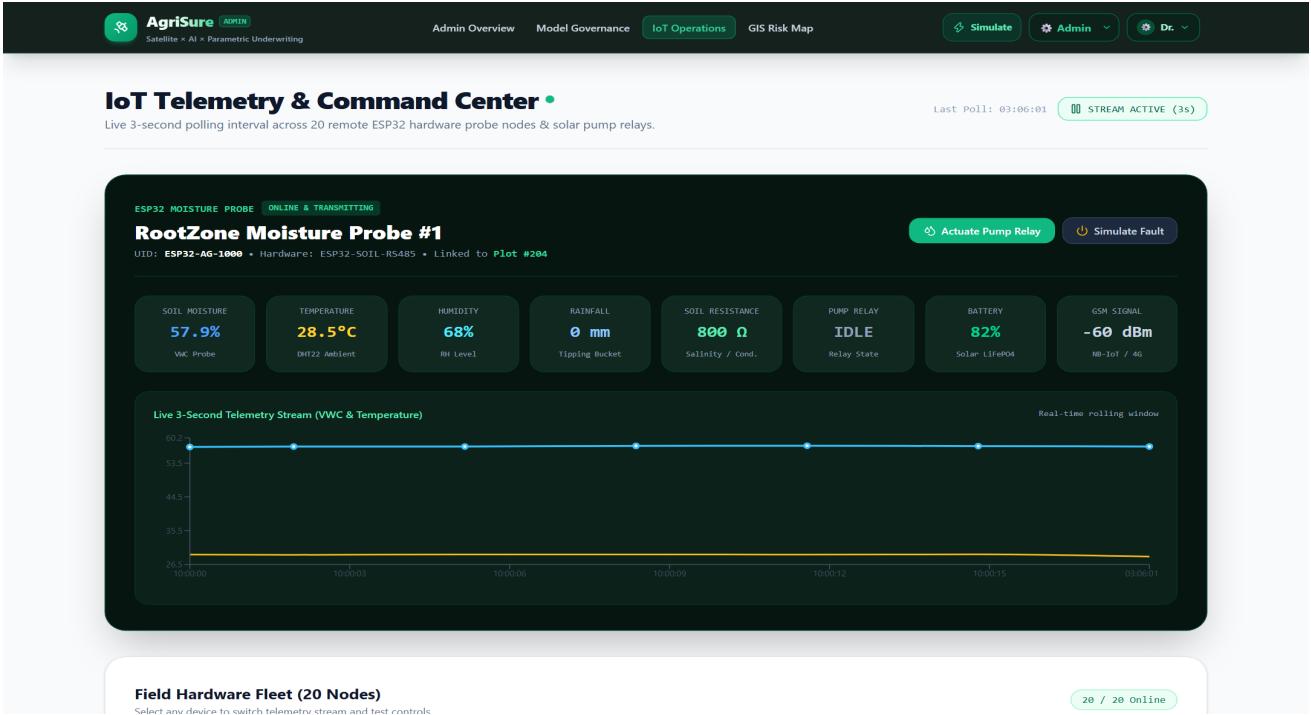
Verification Status	String	Security Badge	No	Chain integrity verification check	CRYPTOGRAPHICALLY VERIFIED ■
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2.16. Hardware IoT Ingestion & Pump Relay Control

Route: /operations/iot

Authorized RBAC Persona: ADMIN & FIELD AGENT | Tier: Hardware Infrastructure Operations

Language Support: English



Functional Purpose & UX Design:

Operations control room monitoring the network of 20 field IoT stations across West Bengal. Displays hardware heartbeat, battery voltage (LiFePO4 4.1V), cellular/LoRa signal RSSI, dual-probe capacitance calibration, and bi-directional pump relay toggles.

- Data Required & Technical Pipeline:**
- **IoT Fleet Endpoint:** GET /api/iot/devices retrieves status and readings across all 20 hardware nodes.
 - **Remote Actuation Dispatch:** POST /api/iot/devices/[id]/toggle pulses hardware GPIO relay on physical device.
 - **Heartbeat Tick Monitor:** POST /api/iot/tick simulates streaming packet arrivals from field edge gateways.

Field Specification & Data Schema:

Field / Metric Name	Data Type	UI Control	Req?	Validation Rules & Technical Logic	Sample Production Value
Device Hardware ID	String (Unique)	Card Header	No	MAC / IMEI hardware device tag	ESP32-NODE-01 (Plot #204)
Installed Plot Location	String	Location Pill	No	Farmland installation site	Baganchra, Santipur Central (Nadia)
Battery Storage Voltage	Float (Volts)	Power Indicator	No	LiFePO4 solar battery voltage: 3.2V to 4.2V	4.12 V (Solar Inflow Active)
Cellular / LoRa RSSI	Integer (dBm)	Signal Bar	No	Signal strength: optimal is > -85 dBm	-78 dBm (Strong Link)
Root Soil Moisture (15cm)	Float (%)	Sensor Metric	No	Volumetric water content in primary root zone	58.0% VWC
Sub-surface Moisture (30cm)	Float (%)	Sensor Metric	No	Deep capacitance moisture reading	64.5% VWC
Soil Temperature	Float (°C)	Sensor Metric	No	Direct in-ground probe temperature	28.4 °C

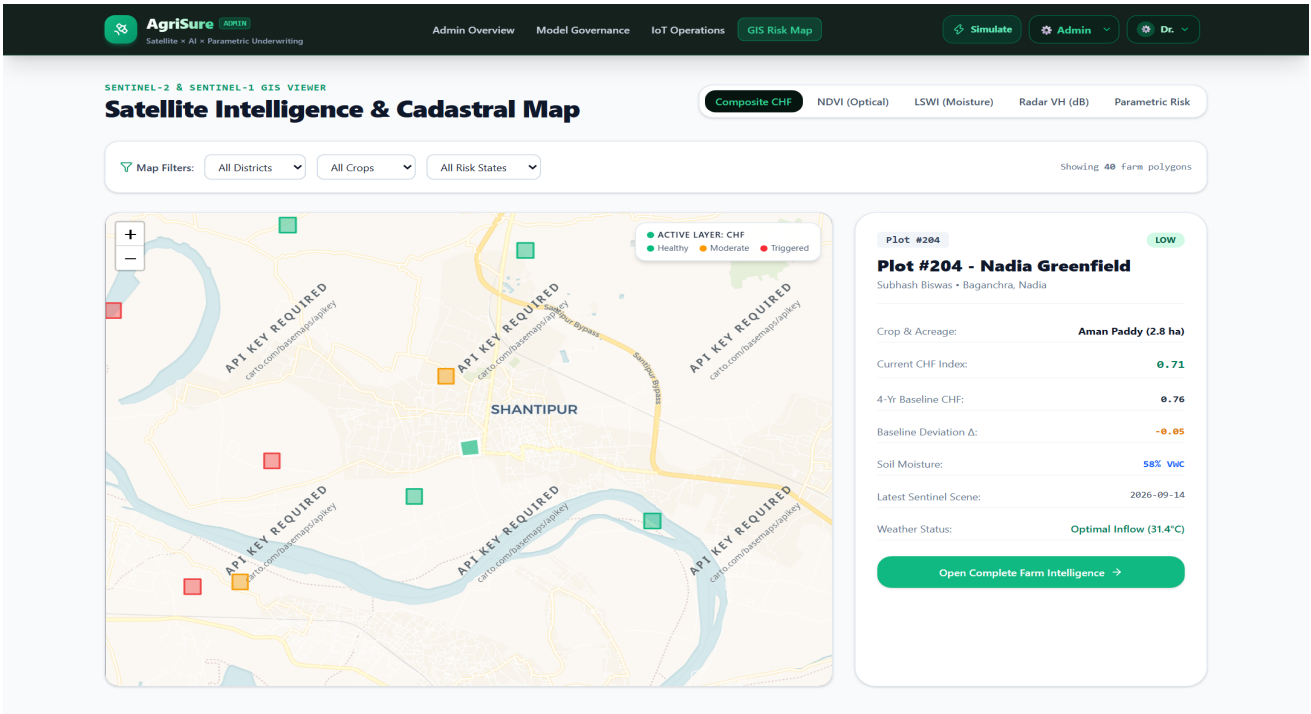
Pump Relay State	Boolean	Toggle Control	Yes	Physical relay contact: CLOSED (1) or OPEN (0)	RELAY ACTIVE (Pump On)
Flowmeter Rate	Float (L/min)	Flow Metric	No	Discharge volume through inline micro-drip line	42.5 L/min

2.17. Sentinel Multispectral GIS Analytics Console

Route: /intelligence/map

Authorized RBAC Persona: ADMIN & FIELD AGENT | Tier: Geospatial Earth Observation

Language Support: English



Functional Purpose & UX Design:

High-resolution spatial GIS explorer overlaying European Space Agency Sentinel-2 optical multispectral imagery and Sentinel-1 C-band SAR radar data. Users toggle false color infrared composites, inspect pixel-level vegetative indices, and analyze cloud penetration.

- Data Required & Technical Pipeline:**
- **Satellite Raster Tile Service:** Streams Sentinel-2 L2A bottom-of-atmosphere reflectance tiles and Sentinel-1 SAR GRD tiles.
 - **Cloud Masking Engine:** Applies Scene Classification Layer (SCL) to filter atmospheric haze and cloud shadows.
 - **Spatial Query Point Intersect:** Allows clicking any pixel coordinate to retrieve full spectral signature (B2-B12).

Field Specification & Data Schema:

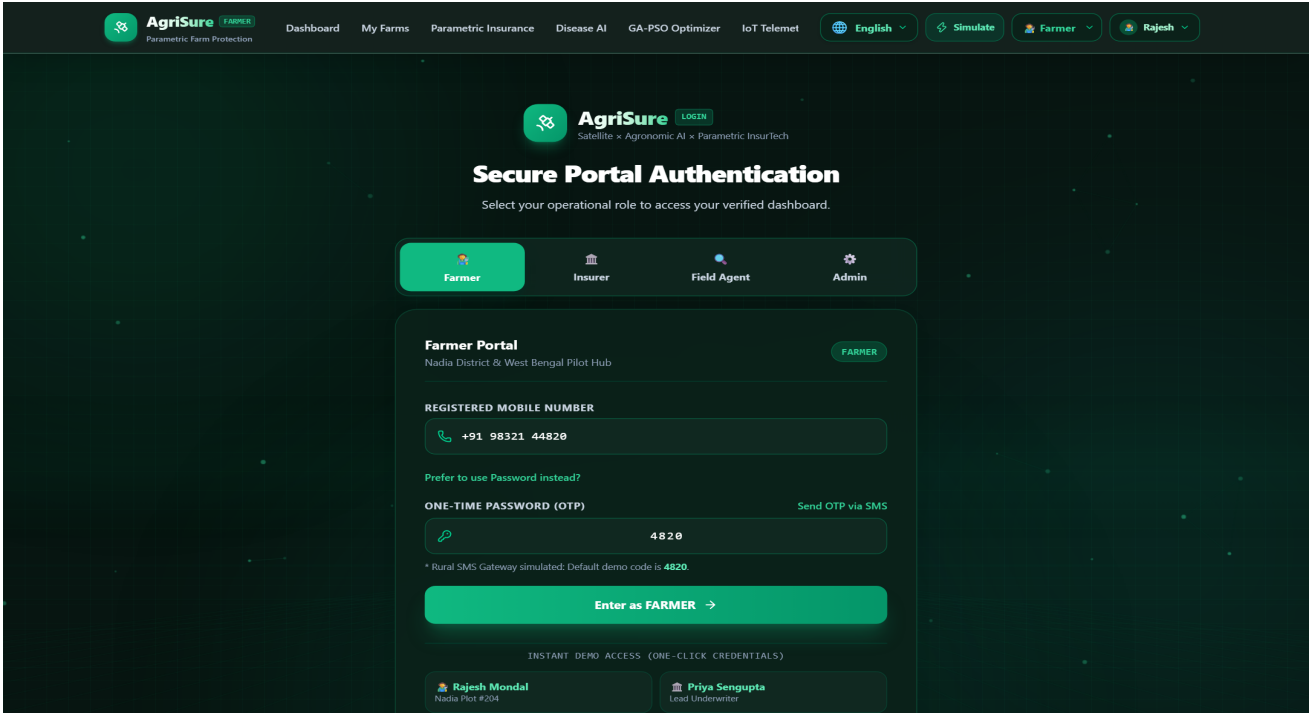
Field / Metric Name	Data Type	UI Control	Req?	Validation Rules & Technical Logic	Sample Production Value
Spectral Band Preset	String (Enum)	Layer Dropdown	Yes	RGB True Color, CIR Infrared, SWIR Agriculture	Agriculture Composite (B11-B8-B2)
Active Remote Index	String (Enum)	Index Selector	Yes	NDVI, NDRE, EVI, LSWI, Radar VH/VV Ratio	NDVI (Normalized Difference Veg)
Cloud Mask Threshold	Integer (%)	Slider Control	Yes	Maximum acceptable scene cloud percentage	15% Maximum Cloud Cover
Pixel Cursor Coordinate	Float (Lat/Lon)	Status Coordinate	No	Geographic location of viewport cursor	LAT 23.472°N • LON 88.556°E
Sensor Satellite Platform	String	Metadata Pill	No	Active spacecraft: Sentinel-2A / Sentinel-2B / Sentinel-1A	Sentinel-2B MSI (10m Resolution)
Pixel CHF Health Score	Float (0.00-1.00)	Inspect Tooltip	No	Calculated vegetative health at sampled point	0.710 (High Vigour)

2.18. Multi-Role Authentication & OTP Portal

Route: /login

Authorized RBAC Persona: Public Universal | Tier: Access Control & Identity

Language Support: English / Role Adaptive



Functional Purpose & UX Design:

Secure entry portal supporting four distinct personas: Farmer, Insurer, System Administrator, and Field Agent. Provides 1-click evaluation presets for instant demonstration, SMS OTP authentication for rural farmers, and credential login for underwriters.

Data Required & Technical Pipeline:

- **Authentication API:** POST /api/auth/login validates credentials, issues auth token, and sets agrisure_user_role cookie.
- **SMS OTP Service:** Simulated OTP dispatch to farmer's mobile number (+91 98321 44820).

Field Specification & Data Schema:

Field / Metric Name	Data Type	UI Control	Req?	Validation Rules & Technical Logic	Sample Production Value
Target Role Persona	String (Enum)	Role Selector Tabs	Yes	Values: FARMER, INSURER, ADMIN, FIELD_AGENT	■■■ Farmer
User Identifier	String	Text Input	Yes	Mobile Number (Farmers) or Work Email (Insurers)	+91 98321 44820 / p.sengupta@agrisure.in
Password / SMS OTP	String (Secret)	Password / OTP Input	Yes	Password or 6-digit one-time SMS verification code	 / 842011
Instant Demo Preset	Button Action	Quick-Fill Buttons	No	One-click autofill of verified persona credentials	Autofill Rajesh Mondal (Farmer)
Keep Logged In	Boolean	Checkbox	No	Extends session cookie lifespan to 30 days	TRUE (Checked)

2.19. 6-Stage Technical Architecture & Agronomic Math

Route: /how-it-works

Authorized RBAC Persona: INSURER & ADMIN | Tier: Technical Documentation

Language Support: English

AgriSureINSURER

Satellite • AI • Parametric Underwriting

Underwriting Desk

Policies Review

IU Risk Map

Claims Desk

Bank Payouts

Audit Ledger

Arc

Simulate

Insurer

Priya

Six-Stage Closed Loop Architecture

How AgriSure Intelligence Works

Click on any stage below to explore its data sources, agronomic principles, and technical pipeline.

01OBSERVE
Satellite Remote Sensing + Field...

02DIAGNOSE
Multi-Stress Detection & Com...

03PRESCRIBE
GA-PSO Hybrid Nutrient & Irr...

04ACTUATE
Automated Hardware Relays &...

05VERIFY
Post-Intervention Satellite Rec...

06INSURE
Objective Parametric Underw...

STAGE 01OBSERVE: Satellite Remote Sensing + Field IoT Probes

Continuous multi-modal observations fusing Sentinel-2 Multi-Spectral Instrument (10m optical) with Sentinel-1 C-band Synthetic Aperture Radar (SAR) and real-time 3-second ESP32 soil probes.

Agonomic & Biophysical Foundation

Radar backscatter (Sentinel-1 VH) penetrates cloud cover and measures structural plant water content before cellular chlorophyll breakdown manifests in visible bands.

Technical Stack & Algorithms

Copernicus Open Access Hub, Leaflet GIS, PostGIS Spatial Indexing

Data Feeds: Sentinel-2A/B (Bands 2,3,4,8,11,12) • Sentinel-1B (VH/VV SAR) • ESP32 RS-485 Soil Moisture Probes • DHT22 Microclimate Stations

LIVE OUTPUT EMISSION LOG

> Plot #204: NDVI=0.68, LSWI=0.38, VH Backscatter=-16.4 dB, Root Moisture=58% VWC

STATUS: VERIFIED

Functional Purpose & UX Design:

Comprehensive technical specification detailing the mathematical foundations: the Shannon Entropy Crop Health Factor formula, GA-PSO bio-inspired heuristic equations, synthetic aperture radar backscatter processing, copula spatial risk aggregation, and the complete 6-stage operational protocol.

Data Required & Technical Pipeline:

- **Mathematical Specifications:** LaTeX-rendered equations for Shannon entropy sub-index weights and GA-PSO fitness functions.
- **Agronomic Protocol Validation:** Citations of university research models benchmarked against West Bengal agro-ecological zones.

Field Specification & Data Schema:

Field / Metric Name	Data Type	UI Control	Req?	Validation Rules & Technical Logic	Sample Production Value
Shannon Entropy Formula	Equation	Math Block	No	$CHF = \sum w_i \times I_i$ where w_i is derived via entropy weights	CHF Formula Matrix
GA-PSO Fitness Function	Equation	Math Block	No	Minimizes cost & nutrient leaching; maximizes vegetative vigour	Fitness Objective Equation
SAR Polarimetric Ratio	Equation	Math Block	No	VH/VV cross-polarization backscatter ratio for soil moisture	$\gamma = \sigma^0_{0_VH} / \sigma^0_{0_VV}$
Parametric Trigger Spec	Rule Spec	Protocol Block	No	Threshold breach specification and claim qualification logic	$CHF \leq 0.580$ for 14 consecutive days

Autonomous Agro-Resilience • Sentinel-2/1 SAR • IoT Telemetry • Parametric Underwriting

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3. Database Schema & Data Dictionary (Prisma ORM)

The system persists all agro-financial state across 22 PostgreSQL models defined in prisma/schema.prisma with PostGIS geometry support. Below is the operational data dictionary of the primary core entities:

Entity Model	Primary Key	Key Backing Attributes	Operational Role in Platform
User	id (UUID)	name, role (Enum), phone, email, district, village, bankAccount	Authenticates Farmers, Insurers, Admins, and Agents with RBAC isolation.
Farm	id (UUID)	farmerId, plotNumber, areaHa, crop, boundaryGeoJson, iuCode	Cadastral plot record with spatial boundaries for satellite intersection.
IoTDevice	id (UUID)	farmId, deviceCode, batteryV, rssiDbm, pumpRelayActive, lastPing	Hardware telemetry ingestion node and remote micro-irrigation pump relay.
SensorReading	id (UUID)	deviceId, soilMoisture15, soilMoisture30, soilTemp, ambientRh, recordedAt	High-frequency 3s timeseries tracking volumetric root zone water content.
CropHealthRecord	id (UUID)	farmId, chfIndex, ndvi, ndre, lswi, sarBackscatter, sourceDate	Computed Shannon entropy composite vigour derived from Sentinel-2/1 passes.
Policy	id (UUID)	policyNumber, farmId, farmerId, sumInsured, grossPremium, subsidyAmount, farmerShare, triggerChf, status	Smart parametric contract binding farmer, underwriter, and index trigger threshold.
Claim	id (UUID)	claimNumber, policyId, perilType, triggerDate, deficitChf, payoutAmount, status, evidenceDossierJson	Automated claim qualification record backed by 11-section empirical evidence package.
Payout	id (UUID)	claimId, farmerId, amount, bankName, ifsc, utrNumber, stage (1-7), settledAt	Direct Benefit Transfer ledger tracking electronic funds settlement to farmer's account.
AuditBlock	id (UUID)	blockNumber, entityType, actorId, prevHash, merkleHash, payloadJson, createdAt	Chained SHA-256 Merkle ledger certifying immutable platform transparency.

4. REST API Endpoint Inventory & Edge Security

All API routes are served from Next.js 16 Route Handlers (src/app/api/*) under strict HTTP method conventions and edge proxy guard verification:

Method	Endpoint Path	Required Role	Data Payload & Response Summary
POST	/api/auth/login	Universal	Receives credentials/OTP; sets agrisure_user_role cookie; returns profile.
POST	/api/auth/register	Universal	Creates farmer or institutional user account; stores initial profile record.
GET	/api/farms	FARMER, AGENT	Returns list of registered farms with GeoJSON boundaries and vegetative scores.
GET	/api/farms/[id]/sensors	FARMER, AGENT	Returns real-time soil moisture (15cm & 30cm), temp, and ambient humidity.
POST	/api/disease/predict	FARMER	Multipart leaf image → 38-class CV inference, confidence, and spray remedy.
POST	/api/optimization/run	FARMER	Soil N-P-K baseline → GA-PSO fertilizer (Urea, DAP, MOP) and water schedule.
POST	/api/iot/devices/[id]/toggle	FARMER, ADMIN	Toggles physical pump relay between ACTIVE (1) and STANDBY (0).
GET	/api/policies	Universal	Lists policies filtered by status (ACTIVE, PENDING_APPROVAL, REJECTED).
POST	/api/policies	FARMER	Submits new insurance application with subsidy calculation and bank verification.
POST	/api/policies/[id]/decision	INSURER	Accepts or rejects application with formal reason and audit ledger inscription.
GET	/api/claims	INSURER	Retrieves all claims with 11-section empirical cryptographic evidence dossiers.
POST	/api/claims/[id]/approve	INSURER	Approves qualified parametric claim and queues it for automated bank disbursement.
GET	/api/payouts	Universal	Lists 7-stage automated bank compensation transfers with simulated RBI UTRs.
POST	/api/payouts/[id]/execute	INSURER	Executes instantaneous electronic direct benefit deposit to farmer's bank account.
GET	/api/audit	INSURER, ADMIN	Streams immutable chained SHA-256 Merkle audit blocks for regulatory compliance.
POST	/api/simulate/event	Universal (Demo)	Injects synthetic weather shock: Severe Drought, Flash Flood, or Foliar Blight.

Deployment Topology & Production Runtime:

- **Vercel Production Edge:** Hosts the Next.js 16 App Router frontend with Turbopack acceleration, regional edge middleware route gating, and optimized static asset delivery.
- **Render Cloud Platform:** Hosts the containerized database layer (PostgreSQL 16 with PostGIS spatial extension) and telemetry background daemon workers.
- **Edge Route Protection (src/middleware.ts):** Inspects incoming agrisure_user_role cookies. Blocks and redirects unauthorized cross-role route attempts (e.g. Farmers cannot visit /insurer/* or /admin/*; Insurers cannot visit /farmer/*).
- **Client-Side Watchdog (RouteGuard.tsx):** Intercepts client navigation and displays an animated modal redirecting the user to their authorized dashboard within 1.2 seconds.